

## M003 Extension — English Worksheet (Intermediate)

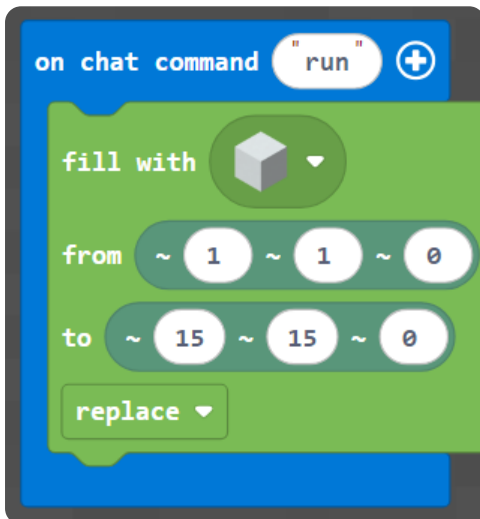
**Topic:** Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Week 3) ·

**Time:** about 40 minutes

This is a **bonus picture challenge** between Week 3 and Week 4. Now you copy a whole mob — a **pink axolotl** — onto your wall. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

**Build your canvas.** Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

### Blocks



### Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

### 1 · Predict 🧠

Read each set of steps. Write a sentence about what you will see.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Three pink blocks share  $y = 13$ . What shape do they make, and which number stays the same?

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```
place black block at (8, 7)
place black block at (8, 8)
place black block at (8, 9)
```

These three share  $x = 8$ . What shape do they make, and which number changes?

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```
place pink block at (6, 10)
place black block at (6, 10)
```

Two blocks aim for the same  $(x, y)$ . What do you see — pink, black, or both? Why?

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## 2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

**Bug A** — This should make the axolotl's two eyes, one at (6, 10) and one at (11, 10). Right now both land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The dragon's eye should sit at (8, 10). Right now the two numbers are **swapped**, so it lands in the wrong place.

```
place purple block at (10, 8)
```

**Write the fixed code:**

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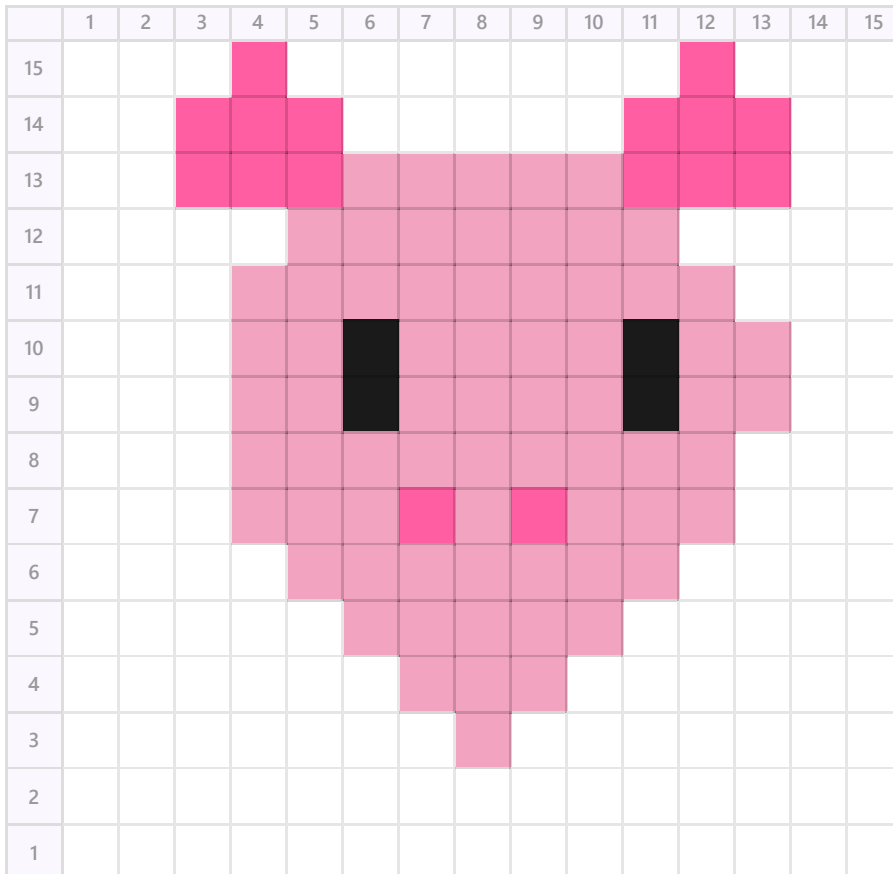
**Why was it wrong? Why does your fix work?**

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### 3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block). Build the 🦎 axolotl below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**; the numbers down the **side** are **y**.

🦎 **Axolotl** (pink · #f2a3c0, frills · #ff5fa2, eyes · black)



Record **one video** (a phone is fine). Show two things:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera. Name the parts.

Fill the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

Write your lines here, then say them in your video.

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Submit 

Send this worksheet + your video to teacher on KakaoTalk.